6-5 Numerical Calculations [1]

Point!

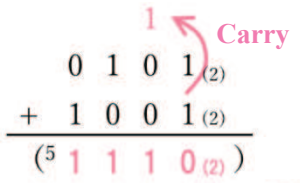
Addition and Subtraction of Binary Numbers

Addition and subtraction in binary numbers are performed digit by digit, just like in decimal numbers.

(1) Binary addition : In binary addition, the place value carries over by one position when the sum reaches 2.

$0_{(2)} + 0_{(2)} = ({}^{1}0_{(2)})$, $0_{(2)} + 1_{(2)} = ({}^{2}1_{(2)})$, $1_{(2)} + 1_{(2)} = ({}^{3}10_{(2)})$, $1_{(2)} + 1_{(2)} + 1_{(2)} = ({}^{4}11_{(2)})$

<Example> Addition of binary numbers $0101_{(2)} + 1001_{(2)}$

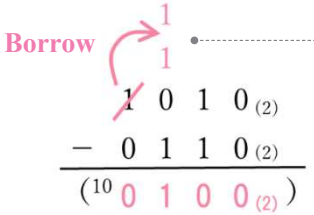


When two 1s are added together, a carry of 1 is generated to the next higher digit.

(2) Binary subtraction : In binary subtraction, when the minuend is insufficient, a value of 2 is borrowed from the next higher digit.

$0_{(2)} - 0_{(2)} = ({}^{6}0_{(2)})$, $1_{(2)} - 0_{(2)} = ({}^{7}1_{(2)})$, $1_{(2)} - 1_{(2)} = ({}^{8}0_{(2)})$, $10_{(2)} - 1_{(2)} = ({}^{9}1_{(2)})$

<Example> Subtraction of binary numbers $1010_{(2)}$ - $0110_{(2)}$



When 1 is borrowed from the higher digit, it becomes two 1s in the lower digit.

Warm Up

Perform the following binary calculations.

(1) $0101_{(2)} + 1010_{(2)}$

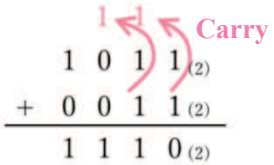
(2) $1011_{(2)} + 0011_{(2)}$

(3) $1011_{(2)} - 0101_{(2)}$

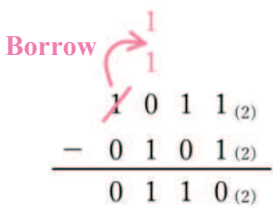
(4) $1101_{(2)} - 1010_{(2)}$

Explanation

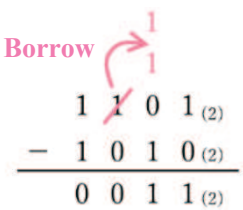
(1)
$$\begin{array}{r} 0\ 1\ 0\ 1_{(2)} \\ +\ 1\ 0\ 1\ 0_{(2)} \\ \hline 1\ 1\ 1\ 1_{(2)} \end{array}$$
Therefore, $\underline{1111_{(2)}}$

(2)


Therefore, $\underline{1110_{(2)}}$

(3)


Therefore, $\underline{0110_{(2)}}$

(4)


Therefore, $\underline{0011_{(2)}}$

Try

Perform the following binary calculations.

(1) $0101_{(2)} + 0110_{(2)}$

(2) $1101_{(2)} + 0010_{(2)}$

(3) $1101_{(2)} - 0010_{(2)}$

(4) $1010_{(2)} - 0101_{(2)}$

Exercise

1 Perform the following binary calculations.

(1) $0011_{(2)} + 1100_{(2)}$

(2) $1001_{(2)} + 0101_{(2)}$

(3) $0111_{(2)} + 0001_{(2)}$

(4) $1010_{(2)} + 0101_{(2)}$

(5) $1101_{(2)} - 1100_{(2)}$

(6) $1011_{(2)} - 0110_{(2)}$

(7) $1001_{(2)} - 0110_{(2)}$

(8) $1010_{(2)} - 1010_{(2)}$